

**Question 1** | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n], dp[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11         dp[i] = 1;
12     }
13
14     int max = 1;
15
16     for (int i = 1; i < n; i++) {
17         for (int j = 0; j < i; j++) {
18             if (a[j] <= a[i] && dp[j] + 1 > dp[i])
19                 dp[i] = dp[j] + 1;
20         }
21
22         if (dp[i] > max)
23             max = dp[i];
24     }
25
26     printf("%d", max);
27
28     return 0;
29 }
```

|   | Input                 | Expected | Got |   |
|---|-----------------------|----------|-----|---|
| ✓ | 9<br>-1 3 4 5 2 2 2 3 | 6        | 6   | ✓ |
| ✓ | 7<br>1 2 2 4 5 7 6    | 6        | 6   | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)